

Real-World Real Estate

Pre-Sale with Proof & Release

FUNDACIÓN BLOCKCHAIN ARGENTINA

Project Catalyst

MILESTONE 4 · EVIDENCE 2

Mainnet Deployment

Mainnet URL and contract addresses, the mainnet transaction list, operational dashboards, and governance and operations playbooks

Project Catalyst 1400106 · Real-World Real Estate: Pre-Sale with Proof & Release · October 31, 2026

Mainnet Deployment

This document is the evidence for **Output 2** of Milestone 4: *"Mainnet deployment with published mainnet URL & contract addresses; operational dashboards; governance/ops playbooks."* Every value in it was read from the live mainnet deployment, the mainnet database and the public Cardano mainnet ledger on **October 31, 2026**, and every transaction was checked against an independent explorer.

What Milestone 4 asks	Where it is answered
Mainnet URL & contract addresses published (Output 2, Criterion 2, Evidence 2)	§1
≥350 on-chain events and ≥100 document hashes (Criterion 2)	§3
Simulated escrow value reported (Criterion 2)	§4
Operational dashboards (Output 2); screenshots of events, wallets, releases (Evidence 2)	§3, §5
Legal greenlight for mainnet from counsel/notary (Criterion 2)	§6
Governance and operations playbooks (Output 2)	§7, §8
List of mainnet TXIDs on a public explorer (Evidence 2)	Annex A, and `mainnet-txids.csv` in the repository

Mainnet has been live since October 31, 2026. The pilot project, Cabildo 4987, was anchored end to end on mainnet the same day: 48 confirmed transactions, every one of them verified on a public explorer, and 0 server errors in the operational dashboards.

At a glance

Web application	https://propnexus-web-mainnet.onrender.com
API	https://propnexus-api-mainnet.onrender.com (base path `/api/v1`)
Network	Cardano **mainnet**
Smart contract	One stage-thread validator, Aiken v1.1.21, **Plutus V3**
Code running in production	commit `fe06df0` of https://github.com/Javote/real-world-real-estate (deploy live October 31, 2026, 20:55 UTC)
On-chain events of the pilot	**48**, all `Confirmed`, between blocks 14,140,863 and 14,141,386
Document hashes committed on-chain	**63** SHA-256 hashes, in 15 Merkle-rooted evidence bundles
Server errors (HTTP 5xx), last 24 h	**0**

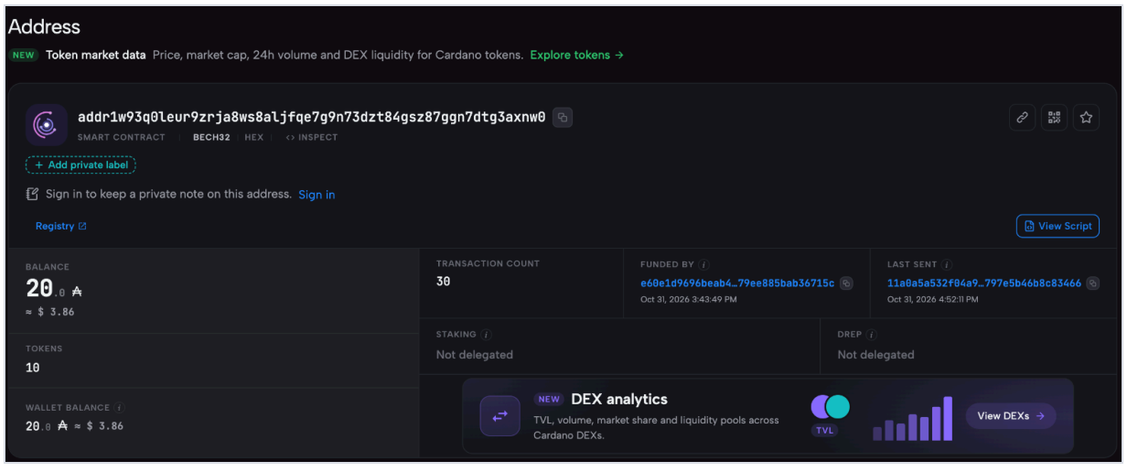
1 • Mainnet URL and contract addresses

Item	Value
Web application	https://propnexus-web-mainnet.onrender.com

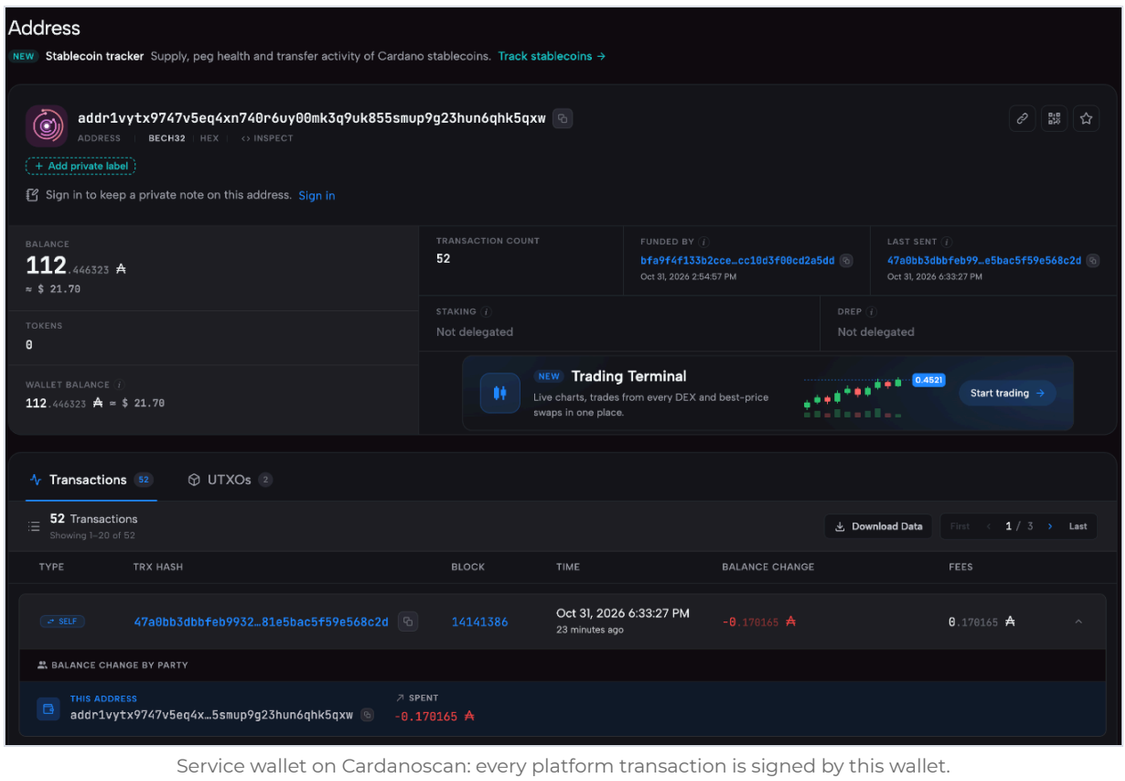
Item	Value
API	https://propnexus-api-mainnet.onrender.com
Script hash / minting policy ID	`6207ff3c19443974ee81fbf92419f20b3f45a259ea8808fe4227e6ad`
Script address	`addr1w93q0leur9zrja8ws8aljfqe7g9n73dzt84gsz87ggn7dtg3axnw0`
Reference script UTxO	`3edf75f533132835612c2d4b03e4268ad20afe0367ffb256a43c2384f3db1ed9#0`
Service wallet (signs every platform transaction)	`addr1vytx9747v5eq4xn740r6uy00mk3q9uk855smup9g23hun6qhk5qxw`
Admin key hash (the validator's only parameter)	`1662fabe65320a9a7eabc7ae11efdda202f2c7a521be04a8546fc9e8`
Compiled blueprint	`contracts/plutus.json` in the repository

How these addresses were verified. None of them is typed by hand in the code: they are derived from the compiled blueprint. For this document they were derived again from `contracts/plutus.json` by applying the admin key hash of the mainnet wallet, and the result matched the published values. The reference script output was checked on the public ledger (Koios): it holds a Plutus V3 script whose hash is exactly `6207ff3c…e6ad`, and it is unspent.

The validator compiled for mainnet is **the same Aiken code that anchored the 180 Preprod events of Milestone 3**; only its parameter (the admin key hash) changes between networks, which is why the mainnet script hash differs from the Preprod one.



Script address on Cardanoscan: 30 transactions, 10 thread tokens (one per stage), 20 ADA held as the protocol minimum that accompanies the tokens.

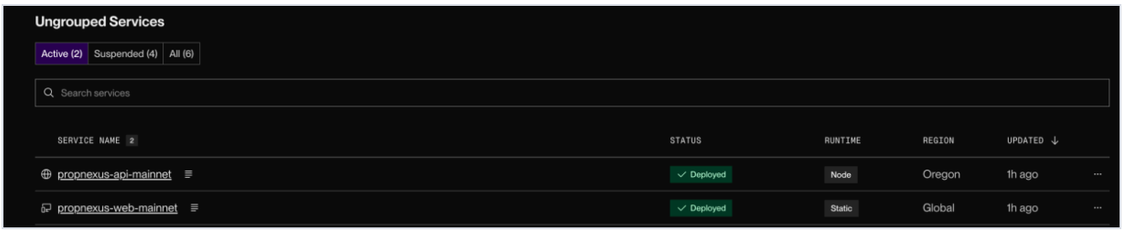


Service wallet on Cardanoscan: every platform transaction is signed by this wallet.

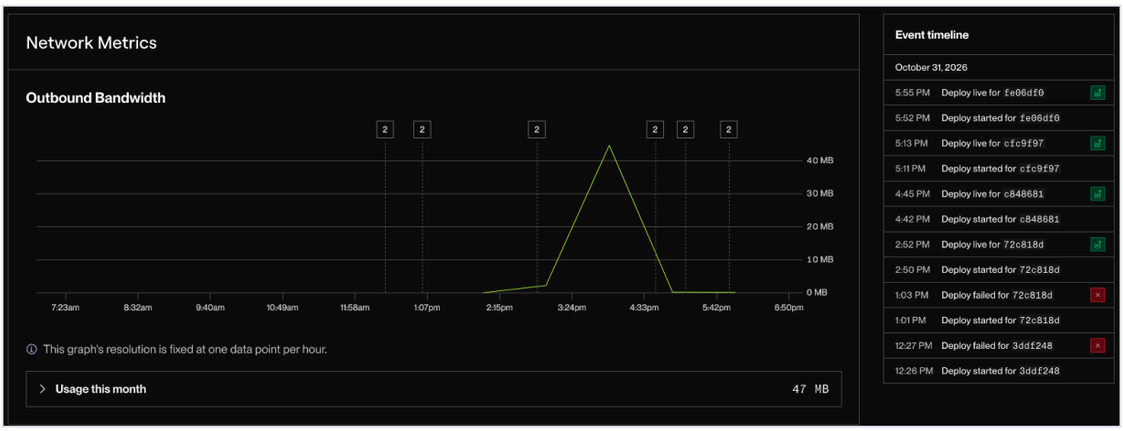
2 · What runs on mainnet

The mainnet environment is a separate deployment from Preprod: its own services, database, file storage, Cardano provider project and wallet. Nothing is shared with the test network except the observability projects (§5).

Layer	Mainnet	Notes
Web	Render static site `propnexus-web-mainnet`	Built with the mainnet API origin and Cardanoscan mainnet as explorer
API	Render web service `propnexus-api-mainnet` (Oregon)	`ANCHOR_MODE=real`, `CARDANO_NETWORK=Mainnet`
Database	Turso `propnexus-mainnet` (`aws-us-west-2`)	Created empty on October 30, 2026, with delete protection
Evidence files	Cloudflare R2 bucket `propnexus-evidencia-mainnet`	Files never leave controlled storage; only their hashes reach the chain
Cardano provider	Blockfrost project `real-estate-mainnet`	Mainnet only
Blueprint	`render.mainnet.yaml`	Mainnet can only be enabled in the `-mainnet` services; never in local development or CI



Both mainnet services deployed on Render.



API service: network metrics and deploy timeline. The last deploy, `fe06df0`, is the code that produced every figure in this document.

3 · On-chain activity

Events and document hashes

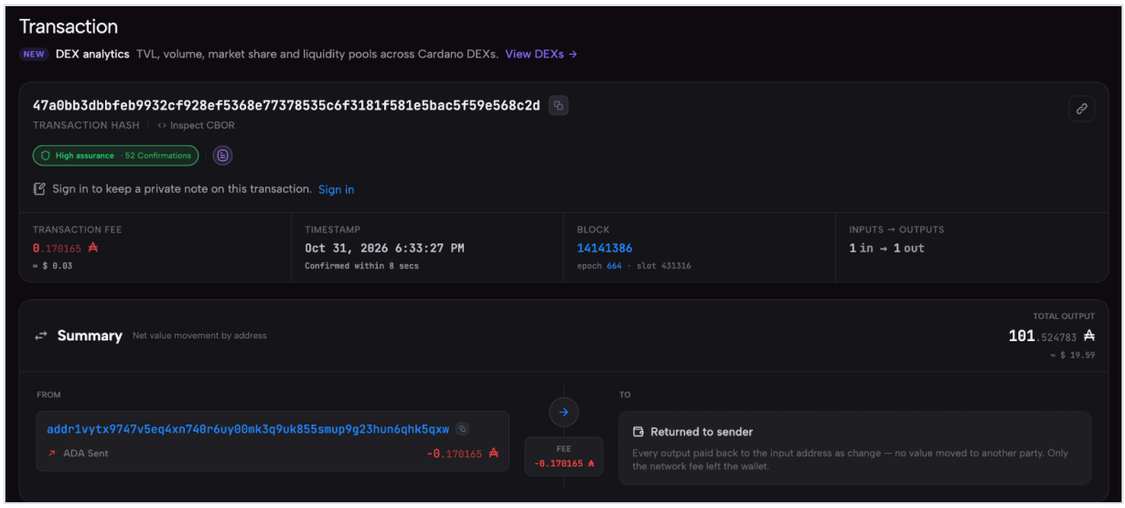
Criterion 2	Target	Achieved on mainnet
On-chain events	≥ 350	**48**
Document hashes	≥ 100	**63**

The pilot ran on a single project (§ *Pilot Report*, Evidence 1), so the volume on mainnet is the volume of one complete project. The targets were not reached and are reported as measured. The throughput of the same code was demonstrated at a larger scale on Preprod in Milestone 3: 180 transactions across 3 projects and 30 stages, all confirmed.

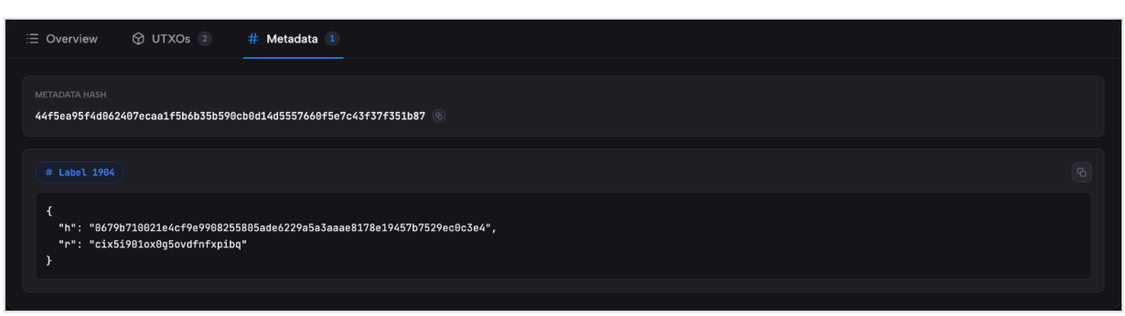
Event type	Count	What it records
Stage thread opened	10	One on-chain thread per construction stage, each holding its thread token
Stage transition	20	`Pending → InProgress` and `InProgress → Completed` for each of the 10 stages
Evidence bundle anchored	15	The Merkle root of a bundle of evidence files, in transaction metadata (label 1904)
Buyer invitation accepted	1	The buyer joining their unit
Dossier signed by notary	1	The notary's signature over the unit dossier
Payment release recorded	1	The record of a payment made outside the platform
Total	**48**	

How each transaction was verified

1. **Existence.** All 48 TXIDs recorded by the platform were looked up on Koios (the community Cardano API, independent of the provider the platform uses): **48 of 48 are on mainnet**.
2. **Commitment.** For the 18 transactions that carry metadata, the hash in the metadata equals the commitment stored by the platform: **18 of 18**.
3. **Evidence bundles.** Each of the 15 bundle Merkle roots was recomputed from the SHA-256 hashes of its files and compared with the root stored and the root on-chain: **15 of 15**, covering all **63** document hashes.
4. **Files.** The 63 stored evidence files (115.7 MB) were downloaded from the platform and hashed again: **63 of 63** reproduce the hash that was anchored.



The payment release transaction on Cardanoscan: the service wallet pays only the network fee; no value moves to another party.



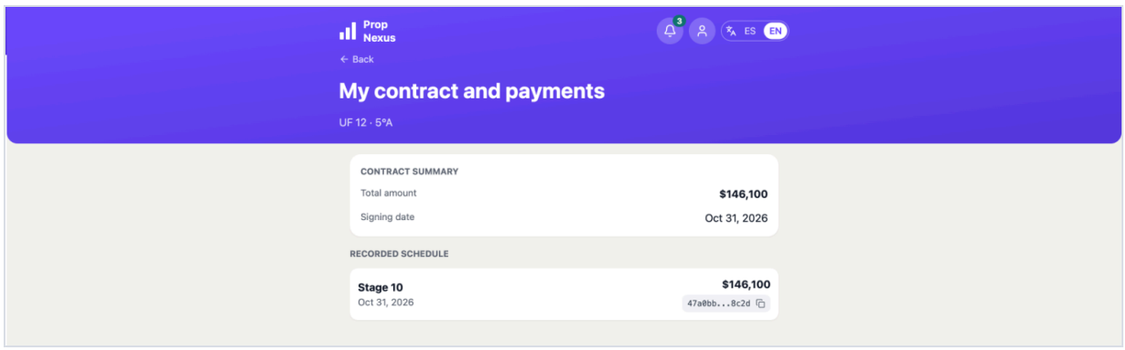
Its metadata (label 1904): `h` is the commitment of the release record, `r` is the opaque identifier of that record. No personal data reaches the chain.

4 · Simulated escrow value

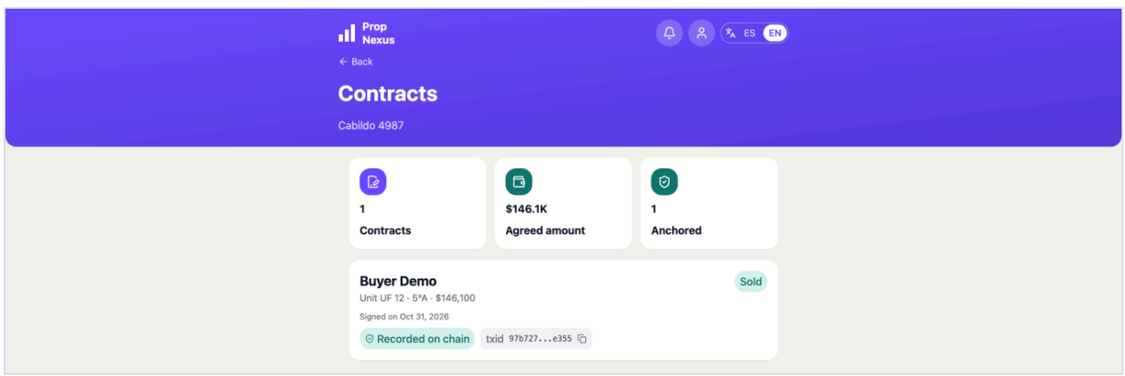
The platform **never holds, transfers or releases funds**: no validator custodies value, and the only ADA locked on-chain is the protocol minimum that accompanies each stage's thread token (2 ADA per stage, 20 ADA for the pilot), paid by the platform's own service wallet. The escrow is therefore *simulated*: the platform tracks the amount a buyer's contract commits, and records each payment made outside the platform, stage by stage, as a hashed and anchored attestation.

	Amount
Value of the contract signed in the pilot (unit UF 12 · 5°A)	**USD 146,100.00**
Payments recorded as released (stage 10)	**USD 146,100.00**
Simulated escrow value still pending release	**USD 0.00**
Value of the 30 units the project holds on the platform	USD 3,665,414.93

Amounts are stored as integer cents, never as floating point. The release is anchored in transaction `47a0bb3d…568c2d` (Annex A, row 48).



The buyer's view of the contract: total amount and the release recorded for stage 10, with its transaction ID.



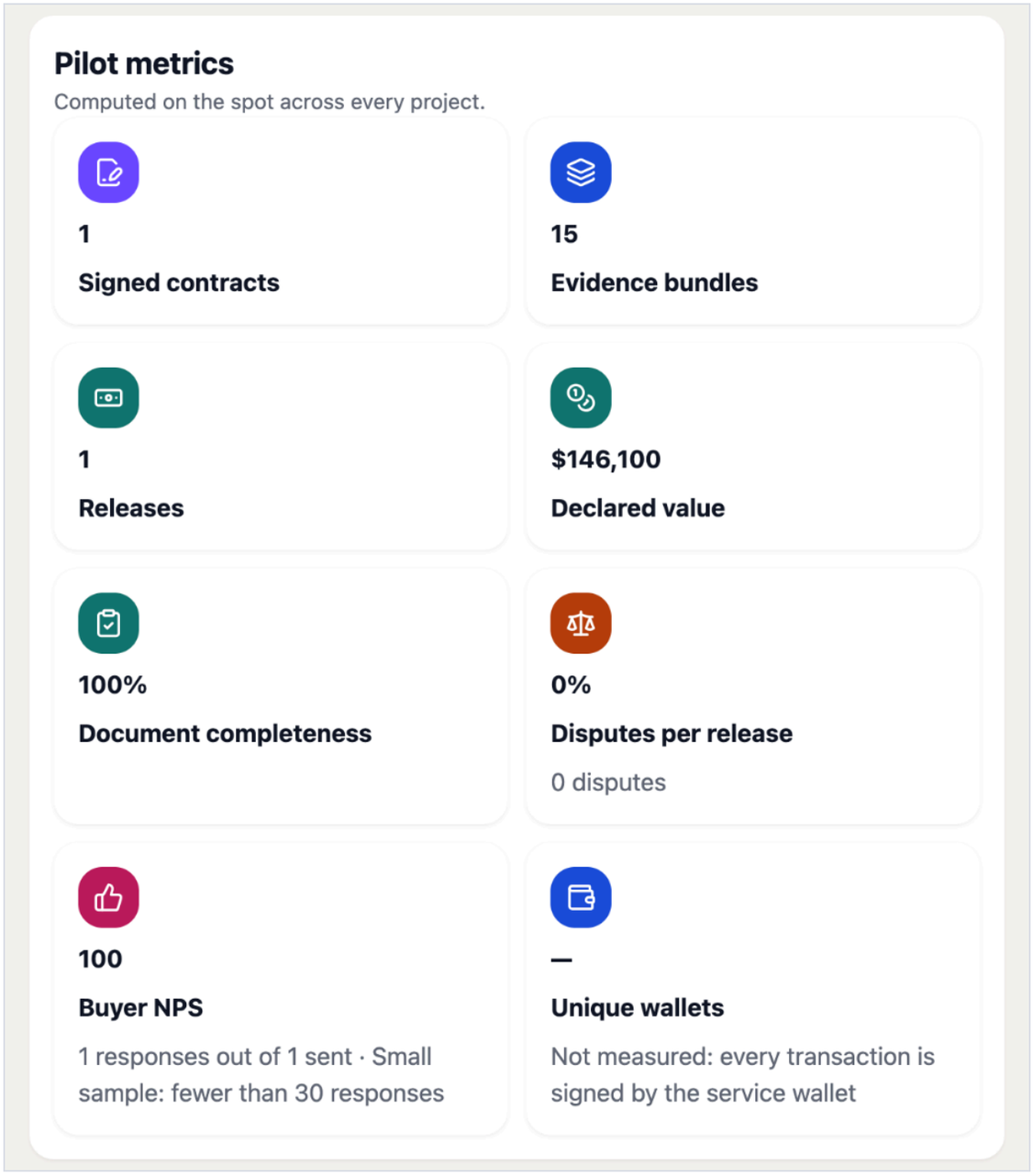
The developer's view: the signed contract, anchored on-chain.

5 · Operational dashboards

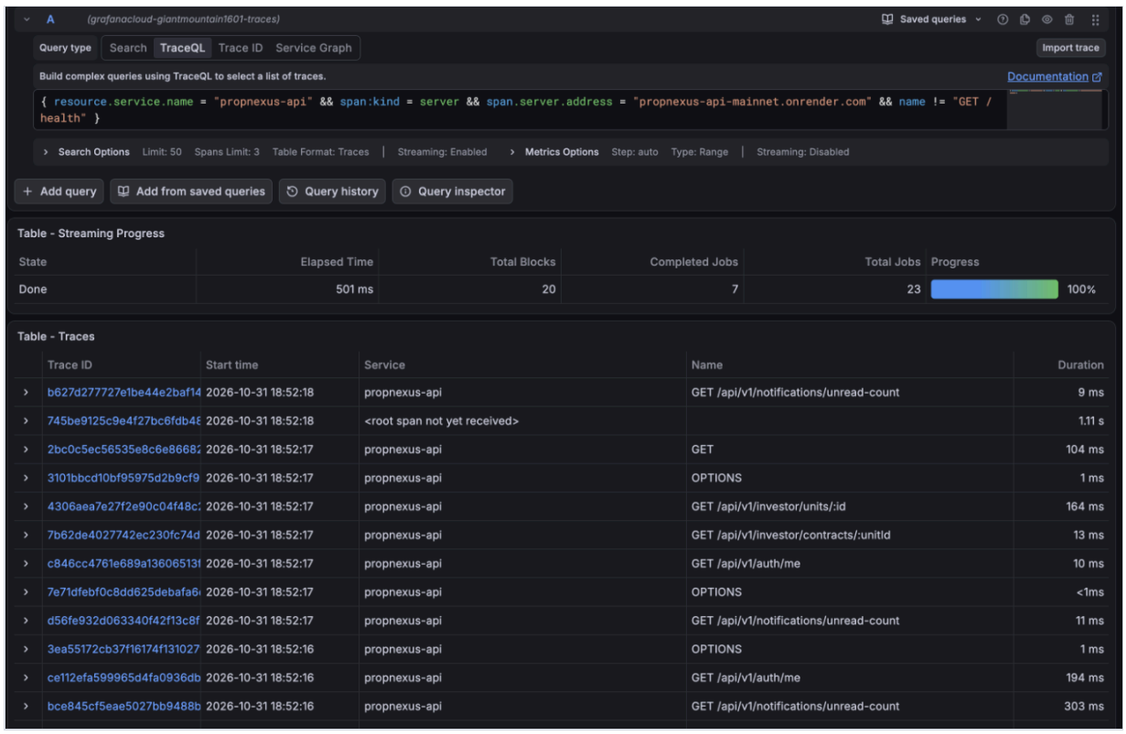
Five tools watch the mainnet deployment, one question each. Sentry, Grafana and PostHog are the same projects used for Preprod (their free plans allow one), so the views below are **filtered to the mainnet hosts**; Render and Blockfrost have mainnet-only projects. The Preprod API has been suspended since October 31, 2026, so it sends no data.

Question	Tool	Mainnet view
What has the pilot produced?	Admin panel of the platform	Pilot metrics, computed on demand

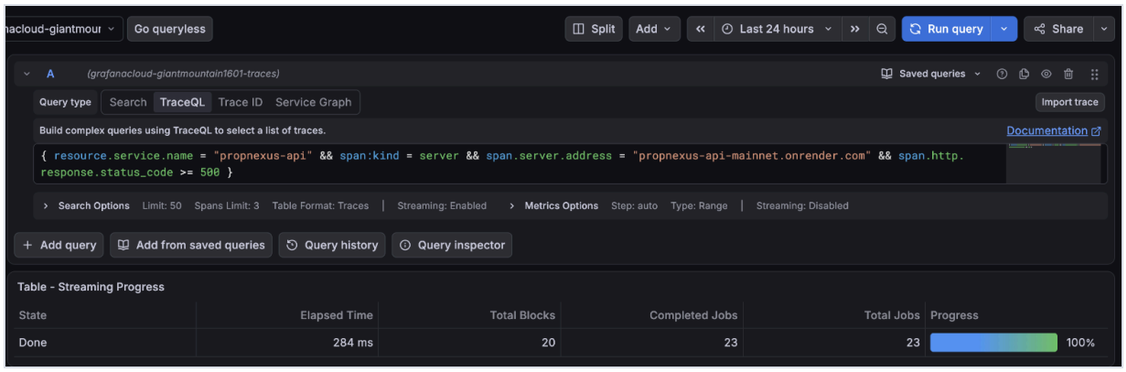
Question	Tool	Mainnet view
How is the server behaving?	Grafana Cloud (OpenTelemetry traces)	Traces filtered to `propnexus-api-mainnet.onrender.com`
Did anything fail for anyone?	Sentry	Issues, every project and environment
What do people do in the app?	PostHog (web analytics, no personal data)	Filtered to host `propnexus-web-mainnet.onrender.com`
Is the infrastructure up?	Render, Blockfrost	Mainnet services and the mainnet provider project



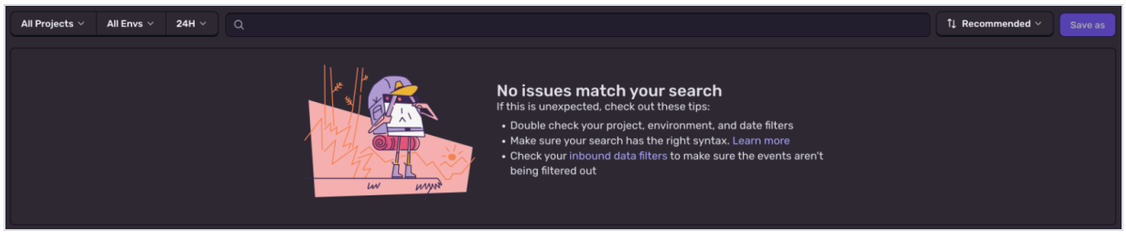
Pilot metrics in the admin panel of the mainnet deployment.



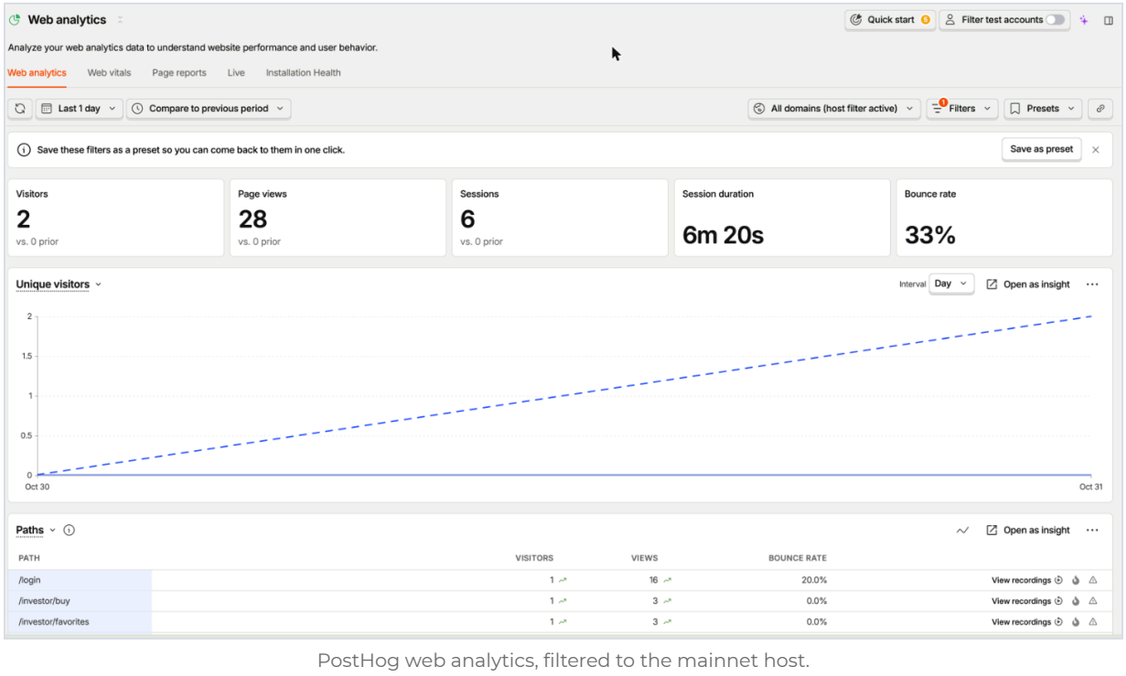
Grafana Cloud: OpenTelemetry traces of the mainnet API (health checks excluded).



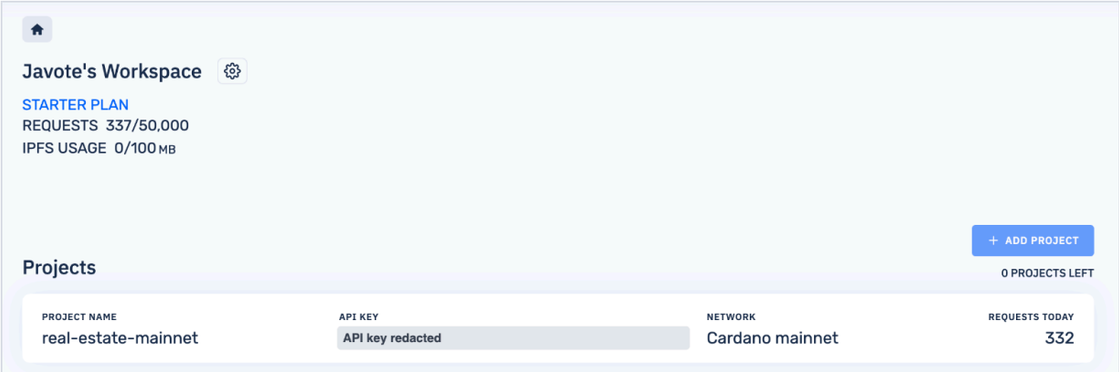
Grafana Cloud: the same traces filtered to responses with status ≥ 500 over the last 24 hours return no results. Error budget consumed: 0.



Sentry: no issues in the last 24 hours, across every project and environment.



PostHog web analytics, filtered to the mainnet host.



Blockfrost: the mainnet provider project (the API key is redacted).

6 · Legal greenlight for mainnet

The legal greenlight for operating on mainnet was granted on **October 23, 2026**, before any mainnet transaction was submitted. Its formal record follows the approval format of the project's Council of Experts (Whitepaper v1.0): a sign-off that lists the reviewed artifacts — the URLs, the script hash and addresses above, the reference script and the service wallet — states what the platform does and does not do, and closes with *APPROVED* and the signature block. It is submitted alongside this document as `M4-Sign-off-Legal-Greenlight-Mainnet.pdf`.

7 · Governance playbook

Governance on mainnet follows the two mechanisms of the Whitepaper — off-chain authorization and on-chain state governance — plus the Council of Experts for approvals, and four operating rules for what only the operator can do.

Who approves what

Decision	Who decides	How it is recorded
A milestone deliverable (whitepaper, pilot plan, evidence pack)	Council of Experts: Developer 1, Developer 2, Notary	Signed e-mail sign-off in the Council format, archived with the milestone evidence
Going live on mainnet	Legal counsel / notary	The legal greenlight sign-off (§6)
Changing code that runs on mainnet	Project owner	A commit on `main`; Render only deploys after the CI checks pass
Changes to contracts, authorization, migrations, storage or deployment	Project owner, line-by-line review	Same, with review before merge
Keys and signing, the service-wallet key, hashing and commitments	Project owner leads; the reviewer must be able to explain every line	Same
Moving the platform back from mainnet to Preprod	Project owner	Change of `CARDANO_NETWORK` in the `-mainnet` services, recorded in the runbook

Roles and permissions in the code

Every request passes **two checks**: the user's global role, and their membership of the specific project. The five roles are developer, buyer (investor), certifier, notary and admin (the platform operator). A buyer's account is bound to their unit; a certifier only sees the stages they were invited to certify; a notary only signs dossiers that are waiting for them. Every relevant change writes an append-only audit-log entry with actor, entity, action and time.

On-chain, the stage state machine is the same one the backend enforces — `Pending → InProgress → {Observed ⇄ InProgress, Completed}` — and the validator only accepts a transition signed by the admin key, carrying a bounded validity window.

Custody of the service-wallet key

- **One payment key, no seed phrase**, generated fresh for mainnet and never stored on a development machine.
- It lives in the API's environment on Render and **offline in two places** kept by the project owner. It is not rotated: the validator is parameterized by its hash.
- The wallet holds only what operations need. Surplus funding is moved out by the owner (as on October 31, 2026: see the control below).

The detection control

A stolen key could anchor false transitions that the validator cannot tell apart. The control is a **weekly comparison of every transaction of the service wallet against the platform's own record**: any transaction the platform does not know is an incident until explained.

Run for this document on October 31, 2026:

	Count
Transactions of the service wallet on mainnet (Koios)	52
Transactions recorded by the platform	48 — all 48 present in the wallet's history
Transactions not recorded by the platform	4, all explained

Transaction	What it is
`bfa9f4f1…2a5dd`	Initial funding of the wallet (2 ADA)
`e448ec8b…a2863`	Funding of the wallet (446.65 ADA)
`3edf75f5…b1ed9`	Publication of the reference script (§1)
`ec5678b2…7ed21`	Owner moving the surplus funding out of the service wallet (305.59 ADA)

8 · Operations playbook

Deploy

1. A change lands on `main`. CI runs lint, type checks, unit tests with 100 % coverage, build, the start-up smoke test, dependency audit, static analysis (Semgrep), end-to-end tests and the Aiken contract checks.
2. Render deploys the `-mainnet` services **only when CI is green**.
3. The deploy is followed until the service is `live` **on that commit**; a request to `/health` wakes the free-tier instance so the deploy is not interrupted.

Rollback

What went wrong	How to go back	What it does not undo
The code	Render → service → Deploys → roll back to the last good deploy (instant)	Anything already written to the database or the chain
The data	Turso point-in-time restore into a **new** database, then point the API at it	Transactions already on mainnet
The network	Owner switches the `-mainnet` services back to Preprod	Mainnet history: it stays on the public ledger, as intended

Nothing on-chain is ever rolled back. A wrong record is corrected by a new, attributable record — which is the point of anchoring.

Incident response on mainnet

Symptom	Likely cause	Response
An anchor stays `Pending` or ends `Failed`	Provider timeout, cold start mid-submission	The record is kept; reconciliation on read and the daily job settle it; a failed anchor is retried from the platform without duplicating it
Anchoring fails for lack of funds	Service wallet below the operating balance	Fund the wallet from the owner's wallet; the anchor stays recorded as `Failed` and is retried once funded
A wallet transaction the platform does not know (detection control)	Owner operation, or a compromised key	Identify it. If not the owner: stop the API, take the key out of Render, inform the pilot participants, and record the incident; existing threads stay verifiable
First request takes about a minute	Free-tier instance asleep after 15 minutes	Expected. Warm it up before a demo or a review
Errors in Sentry or 5xx in Grafana	A defect	Triage from the trace; fix forward on `main`, or roll back the code
Evidence storage near its quota	Large files	R2 free tier: 10 GB per account; the pilot used 115.7 MB

The full deploy, rollback and incident runbook of the platform (Milestone 3) and the mainnet cutover runbook are in the repository; Evidence 3 summarizes them.

Annex A · Mainnet transaction list

Every transaction of the pilot, in the order it was recorded. Each ID links to Cardanoscan; the same list, with the commitment of each event, is in the repository as `specs/evidencia-m4-fuentes/2-mainnet/mainnet-txids.csv`.

#	Event	Stage	Transition	Block	Time (UTC)	Transaction ID
1	Stage thread opened	1	→ Pending	14140863	2026-10-31 18:43:49	e60e1d9696beab44c991ed7f31745f47902c68db8be6a3d979ee885bab36715c
2	Stage thread opened	2	→ Pending	14140863	2026-10-31 18:43:49	57d68715c99a9e357141c2f5a3aba3fb832d19ef77164b71e453fe35db2c7d33
3	Stage thread opened	3	→ Pending	14140863	2026-10-31 18:43:49	7c5a76ad07cf314153454d23b8b2014a99e3ad4d386072affbf8699556037c9d
4	Stage thread opened	4	→ Pending	14140863	2026-10-31 18:43:49	3f29cb027828966ef5c542a2ba72ae9d14a2e93198e8795dc58b597f4b00f482
5	Stage thread opened	5	→ Pending	14140863	2026-10-31 18:43:49	5334e4e88dfcb1f8e5e34e558ebff34d65a7ac6970f68ba28964b51756a3ace4
6	Stage thread opened	6	→ Pending	14140864	2026-10-31 18:43:57	f939074a67034299b3b05adfee0d7d2b53e6cde34365feecb6fc3b548651fca7
7	Stage thread opened	7	→ Pending	14140864	2026-10-31 18:43:57	fc8d237b8516ecc89581ab0b2051208dbd7d8e08fec1bdbefa98b671905abee1

#	Event	Stage	Transition	Block	Time (UTC)	Transaction ID
8	Stage thread opened	8	→ Pending	14140864	2026-10-31 18:43:57	ea5e364f110911f50a5221a3fcbd6687c5a3df7653b982cb974f8f5cf6755a6b
9	Stage thread opened	9	→ Pending	14140864	2026-10-31 18:43:57	d4474a27062e7d763d4c11bbf4e8baae59b13e9956bfa1925d7e0b74cf62c6bd
10	Stage thread opened	10	→ Pending	14140864	2026-10-31 18:43:57	d68c2831e7a63c3bd40bbf99e01af05fa205c39ad29dca20bcd2b03d5a9e715e
11	Evidence bundle anchored	1		14140940	2026-10-31 19:08:18	125b0d580e95f12aea5f7ff69b9914d1d23eb7a5249abe16473abd53691f51e5
12	Stage transition	1	Pending → InProgress	14140940	2026-10-31 19:08:18	9bb6a6762c9c11f4c5f65b25ee1fb9a9d02a646e466b65283f076db4486ee172
13	Evidence bundle anchored	2		14140969	2026-10-31 19:17:22	2daf3acc818437d17a671068b9b8e6329c06d989cf354c5154683e319a6d52ef
14	Stage transition	2	Pending → InProgress	14140969	2026-10-31 19:17:22	c60445ac3456c478e71a969288b535ac1c32490a2dbfb4ffcde7926efc3f5e01
15	Evidence bundle anchored	3		14140974	2026-10-31 19:18:48	1deda3b749e064be09766c1eecdff37e0ebcfc90fbdd29c65bc4bc56269ab67e
16	Stage transition	3	Pending → InProgress	14140974	2026-10-31 19:18:48	3f06e500acdbd7049a276eacc9e61dfbd3dbda17c76678ce3d3d621676ad28c5
17	Evidence bundle anchored	4		14140979	2026-10-31 19:20:01	953912d60b0d40044cedb3ca1e756fb457193d708a3992d1d3192edf30da053c
18	Stage transition	4	Pending → InProgress	14140979	2026-10-31 19:20:01	fcd03a7667d0f6d68a91fe47925c1a8b7345ac048e4b4f44c39388757afc47f2
19	Evidence bundle anchored	5		14140980	2026-10-31 19:20:56	c300c92b7a45abdecbea570df8b8e443acc414c6dd3f54ddd6f4c28e41e99351
20	Stage transition	5	Pending → InProgress	14140980	2026-10-31 19:20:56	770066296995d1e3dedb5ca14d110339d2f3f079ed3df255851fc59e070a42f9
21	Evidence bundle anchored	5		14140983	2026-10-31 19:22:26	0126c5b5111802e26956f69b268a3a60cc4c84fdbcb9c914a9835708a9d22b65
22	Evidence bundle anchored	6		14140985	2026-10-31 19:23:43	062b5cf5ec006bc50d536f4b0b16b183bb486189e108051aea7967395c3a044c
23	Stage transition	6	Pending → InProgress	14140985	2026-10-31 19:23:43	a567dfb8a6520d6a621d2c5ad988998c0b7e48a17da2af0c7af829cdc30595c1
24	Evidence bundle anchored	6		14140995	2026-10-31 19:24:47	c6e919e9dd4e93b5b41b1b43423b967043d184b9386ddac6799edace926d8e86
25	Evidence bundle anchored	6		14140997	2026-10-31 19:25:52	8466fae4b7c4b223e91b244e99ba503deb1e96b6336fa276befcb07a4b63ffdb
26	Evidence bundle anchored	7		14141000	2026-10-31 19:26:59	578a58fdf8743aab97414c242d72d40f9e5c002f2e4a8dceadf85e219d9a61e1
27	Stage transition	7	Pending → InProgress	14141000	2026-10-31 19:26:59	678af2c23affffbc8b2af139485fea3b86a7a0a55747b121967f9912dbf83917
28	Evidence bundle anchored	8		14141001	2026-10-31 19:27:37	cf61dbf16be06a2511c2e34f8ca8bc398f96f5bac0bf2965838d78c7c955e959
29	Stage transition	8	Pending → InProgress	14141002	2026-10-31 19:28:10	8da88ac22508725091d6872eeb3cbfb81f2426f8c8eaaa690ac305dfebd2460a
30	Evidence bundle anchored	9		14141007	2026-10-31 19:29:04	7c35d6e76a6990e5a1c8463161f94cb5a932031958f37ee502ca3819866d5188
31	Stage transition	9	Pending → InProgress	14141007	2026-10-31 19:29:04	516388051482f4e4cabc3bb6f9b55492029cafa2dc3451af4e03a06995defc51
32	Evidence bundle anchored	10		14141011	2026-10-31 19:29:58	c691051293c0a57432f54c06de63e7b32717e99bfdd673c5048c0aa558f3e75a
33	Stage transition	10	Pending → InProgress	14141012	2026-10-31 19:30:00	1f2f541759a6fabd77835eb1b5b7b593cf5c7ce2205ae623a601717b3cb66cee
34	Evidence bundle anchored	10		14141014	2026-10-31 19:31:17	6ec675b9dd37cf9340198f4a4a5d3522f47574f5a753ff4dd9b1ce7a5bfc1428
35	Evidence bundle anchored	10		14141019	2026-10-31 19:32:55	fef6e524c53710d9d508aa04d4aeeb35b9b08a1662657c9e39152c603008e314
36	Stage transition	1	InProgress → Completed	14141046	2026-10-31 19:40:55	591273c7727ef9c80e1ee3d7a94dbbf8708b374b405d7e0526724574f54b4269
37	Stage transition	2	InProgress → Completed	14141050	2026-10-31 19:43:08	dfad5a0ba41b4d93ce39b1513eddf91166e58543f1ad702c56e019daaa61b468
38	Stage transition	3	InProgress → Completed	14141055	2026-10-31 19:43:59	f8cb4f71b9ebf6bcecfc1d74a452677f2967ce3c558b9431359e1a7787aba3f5
39	Stage transition	4	InProgress → Completed	14141061	2026-10-31 19:46:23	9a549bfec91123afedcc4d473a67386607b3aa175ced70558b15274b474d660c
40	Stage transition	5	InProgress → Completed	14141063	2026-10-31 19:47:39	07ced4eea22021f97ccfb6b0b4fffb8b0fe4aa16ee98239d09b14b00384b7558
41	Stage transition	6	InProgress → Completed	14141065	2026-10-31 19:48:50	262bb75c262b74c8f41b14685d9f9a85898f66c0ba8a5520ca3bd5d6f536cac1

#	Event	Stage	Transition	Block	Time (UTC)	Transaction ID
42	Stage transition	7	InProgress → Completed	14141068	2026-10-31 19:49:42	a6ea8cde9195a90f1ba4acebd78ecc5c172f380a0725f3818447cbaed14f176e
43	Stage transition	8	InProgress → Completed	14141071	2026-10-31 19:50:29	35fd5682499111d4d6dc99bc83419da21c9a53491c59486531ed5c05f4c5831e
44	Stage transition	9	InProgress → Completed	14141073	2026-10-31 19:51:22	5988ab38e6301210097094da6170a1abaf316a5077963b56fc07ea438b33f9cc
45	Stage transition	10	InProgress → Completed	14141075	2026-10-31 19:52:11	11a0a5a532f04a9042943aefa2c30a72cdb9813258c88f69797e5b46b8c83466
46	Buyer invitation accepted			14141156	2026-10-31 20:21:28	97b7274f96994de7ad33bffd75bd5d415cb5afe2f18f2bb136ef0b73b5e5e355
47	Dossier signed by notary			14141159	2026-10-31 20:22:21	af744e52c9ef141e61fd8c6da9609cf6f51f57e026bc06bfe7d778f1b376ea20
48	Payment release recorded	10		14141386	2026-10-31 21:33:27	47a0bb3dbbfeb9932cf928ef5368e77378535c6f3181f581e5bac5f59e568c2d